

Bacterial Genome-Wide Association Studies (BGWAS)

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DeGAIN Workshop: Module 2

Outline

- Introduction to BGWAS and application
- The process of BGWAS analyses
- Confounding factors, and approaches to correct for the effect of confounding factors
- Tools and Pipeline(s) for BGWAS

Introduction to GWAS

- **GWAS:** a transformative approach for investigating the **genetic variations** underlying diverse **bacterial phenotypes** at the population genome level (Yang et al., 2025).
- **Genetic variations:** SNPs, genes, indels, unitigs, k-mers.
- **Bacterial phenotypes:** bacterial drug resistance, pathogenicity, host specificity, biofilm formation, probiotic fermentation characteristics.
- **Concept of GWAS:** 1990s, proposed by Risch and Merikangas, genetics of complex diseases. E.g., to identify genetic variants associated with alzheimer's disease, cancer, T2D, coronary artery disease.

Introduction: History and Application

- BGWAS: Sheppard et al (2013) pioneered the application of GWAS in *Campylobacter* – vitamin B5 as a host specificity factor.
- Difference between GWAS and BGWAS
 - Human/Animal reproduction different from bacterial reproduction
 - Bacterial genomes tend to be shaped by stronger positive selection, stronger linkage disequilibrium (LD), and stronger population stratification than humans.
- Application of BGWAS
 - AMR surveillance
 - Vaccine development
 - Precision microbiology

Introduction: Genetic Variation

- SNPs: A SNP is the simplest type of genetic variation: one nucleotide differs between genomes.
- Genes: A complete functional DNA sequence encoding a protein or RNA.
- Kmers: A short DNA sequence of fixed length (k nucleotides) extracted from a genome.
- Unitigs: A sequence of length k derived from many overlapping k -mers

Phenotypes: Bacterial Drug Resistance

- A comprehensive understanding of the mechanisms underlying AMR is crucial for reducing and controlling resistance.
 - *K. pneumoniae*: 28 and 37 potential marker genes associated with imipenem and meropenem resistance (Pei et al., 2022)
 - *S. aureus*: MprF P314L and L826F associated with daptomycin resistance (Weber et al., 2021).
 - *E. coli*: mutations in *bdcA* and *valS* genes implicated in quinolone resistance (Malekian et al., 2022).
- Functional validation through wet-lab experiments remains crucial to confirming their biological relevance.

Phenotypes: bacterial pathogenicity

- The pathogenic potential of a bacterial strain is a multifaceted trait dependent upon the complement of core genome mutations and accessory genes present in that strain.
 - *E. coli*: papGII gene, a major risk factor for urosepsis (Cuénod et al., 2023)
 - *Streptococcus*: genetic variation within the capsule biosynthesis locus, associated with neonatal group B Streptococcus disease onset time and meningeal invasion (Chaguza et al., 2022).
 - Drug-susceptible *M. tuberculosis*: G559D mutation in the virulence gene *otsB1* associated with poor treatment outcomes (Chen et al., 2023).
- Most of these significant genetic variants remain experimentally unvalidated.

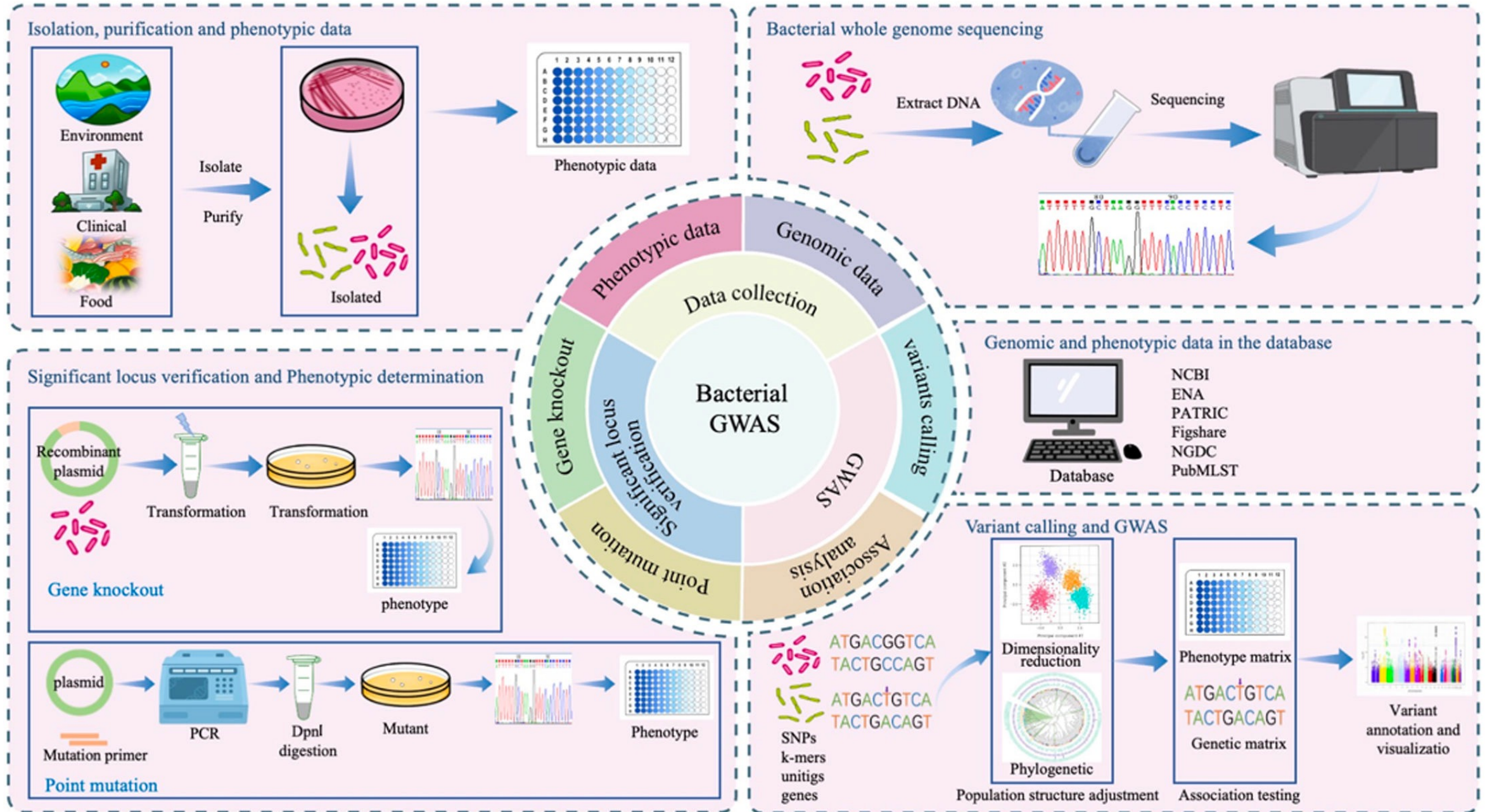
Phenotypes: bacterial host specificity

- Certain genetic factors in bacteria determine their capacity to infect specific hosts or exhibit a preference for particular hosts.
 - *Campylobacter*: 3 genes encoding vitamin B5 biosynthesis almost universally present in isolates from cattle but absent in isolates from chickens and wild birds (Sheppard et al., 2013).
 - *Salmonella enterica* subsp. *enterica*: 52 genomic signatures associated with isolates from avian, bovine, swine, and fish sources (Vila Nova et al., 2019).

Other bacterial phenotypes

- Bacterial biofilms: facilitate bacterial colonization and significantly enhance resistance to antimicrobial agents and host immune responses.
 - *P. aeruginosa*: a 165 kb genomic island (containing heavy metal resistance genes, transcriptional regulators, methyltransferases). associated with enhanced early biofilm formation at 22 °C compared to 37 °C (Redfern et al., 2021).
 - *L. monocytogenes*: 273 genes associated with enhanced biofilm-forming capacity (Monteith et al., 2023).
- Further work e.g., gene-knockout studies is required to confirm their relevance.

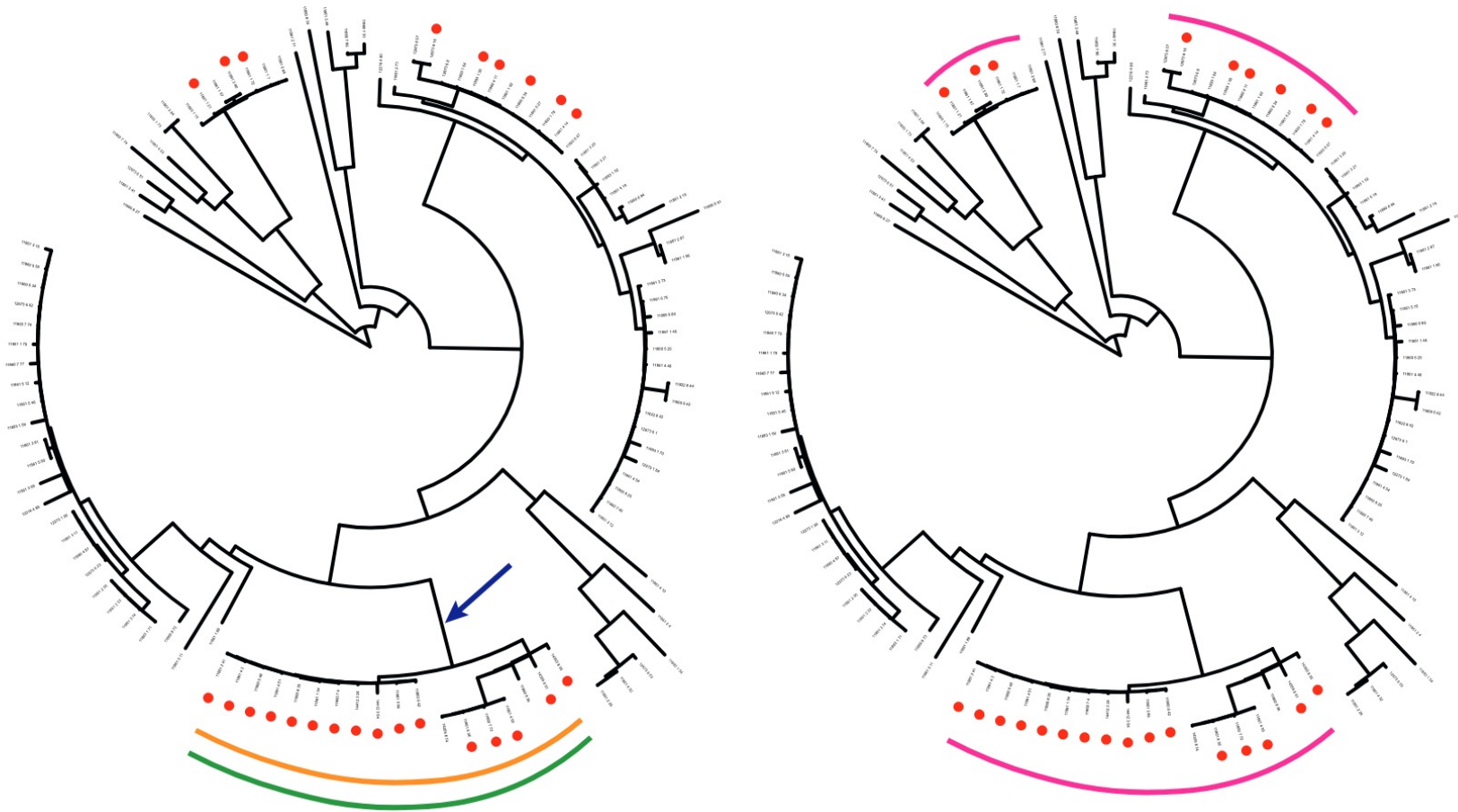
The Process of BGWAS analysis



The Process of BGWAS analysis

- Things to consider
 - Phenotype selection: traits should be clearly **distinguishable** and readily **measurable** to ensure accurate association analysis.
 - Genotype data acquisition: WGS and/or download data from databases (NCBI, ENA, etc)
 - Method/approach for association analysis (BGWAS), dependent on the type of genetic variation, specific phenotypic traits under investigation. Choice of genomic markers (k-mers or unitigs are recommended).
 - Validation of significant loci: Gene knockout, point mutation experiments.

BGWAS confounding factors: Population Structure



Source: Lees, 2017

Figure 1: Phylogenetic illustration of lineage and locus variants. Depicted is an example phylogeny, with cases identified by red dots at the tips, and controls without dots. Variant presence is shown as coloured arcs. a): The yellow variant is a causal lineage variant, and will be associated with the phenotype in a naive analysis. However the green variant, present in the same clade, is not causal but will also appear associated at the same level of significance. Indeed, any mutation that has occurred on the branch indicated by the arrow will appear associated, hindering association mapping. b): The magenta variant has arisen independently in three separate clades containing cases, giving more independence from genetic background and more evidence for association with the phenotype. The association should have a higher p-value, and slightly lower odds-ratio (OR) than the green and yellow variants due to the reduced penetrance observed.

If not corrected, this would lead to

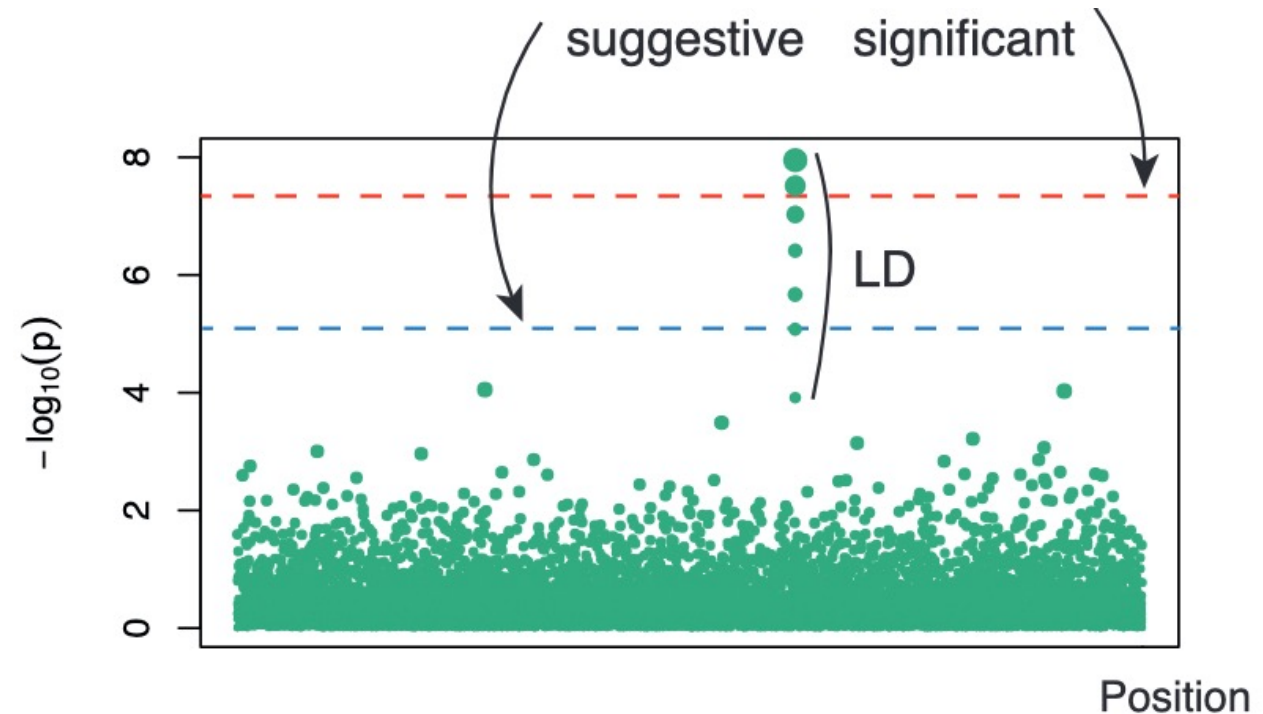
- false-positive associations that enter the literature and require subsequent retraction or correction.
- Missed causal variant: True associations can be masked by strong lineage effects.

Approaches to correct for the effects of population structure

- Clustering strains and subsequently conducting association testing within these subgroups (e.g., using fastBAPs).
- Dimensionality reduction techniques (PCA and MDS).
- Phylogeny-based correction methods, to correct population structure biases due to lineage relatedness.
- Inclusion of random effects in the mixed models to account for genetic relatedness among the strains (Imm mode in pyseer).

BGWAS confounding factors: Linkage Disequilibrium (LD)

- Linkage disequilibrium (LD): It refers to the non-random association of alleles at different loci within individuals in a given population.



Plot all p-values on a log scale versus their position, those exceeding multiple testing adjusted significance thresholds are 'hits'. LD between nearby sites will make multiple variants appear associated in a locus, though usually only one is causal.

Approaches to correct for the effects of Linkage Disequilibrium

- Expanding the number of strains (sample size). This increases the chances of observing independent evolutionary events.
- Condition GWAS study on known resistance mutations (Ma et al., 2020). Conditioning removes the variation already explained by the known mutation. Example on *gyrA* mutation vs Efflux pump mutation.

Difference between Population Structure and Linkage Disequilibrium (LD)

Population structure asks:

"Is this variant associated because it belongs to a particular lineage?"

LD asks:

"Is this variant associated because it is physically linked to the causal variant?"

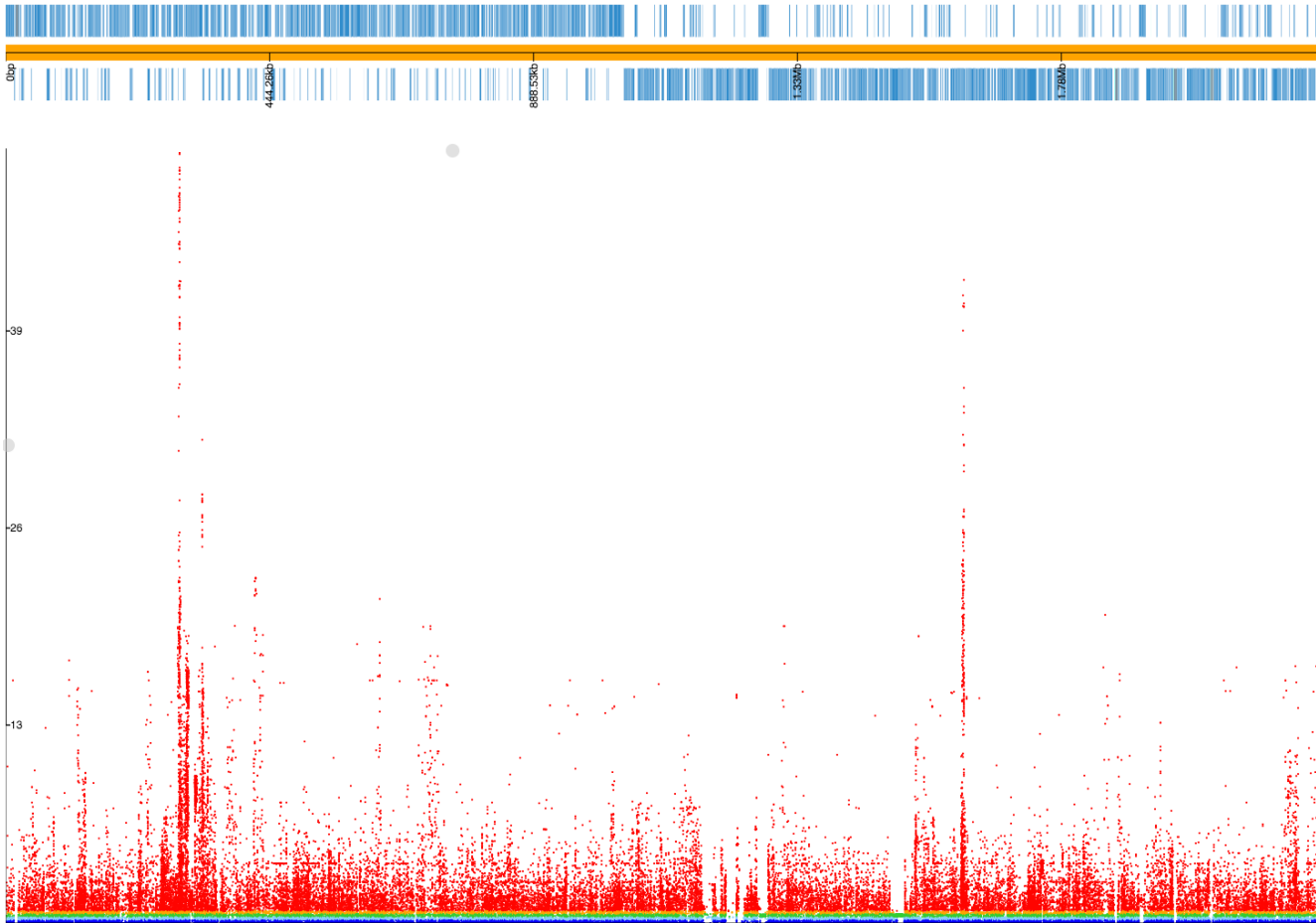
BGWAS confounding factors

- Multiple testing: multiple independent association test for each genotype and phenotype of interest. This can be corrected by
 - Statistical corrections such as Bonferroni, Benjamini-Hochberg, etc
 - Reducing the number of tests.

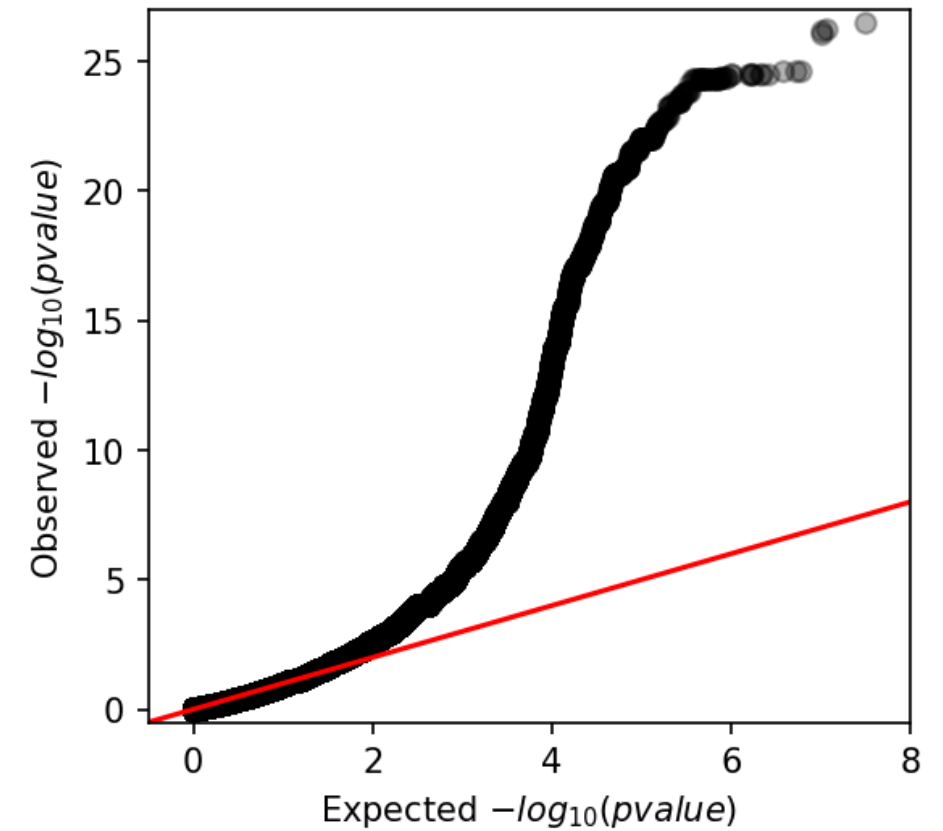
Tools and Pipelines for BGWAS

- Dependent on study aims. Eg., the specific phenotypic traits under investigation, the type of genetic variation, sample size.
- Tools include pySEER, scoary2
- Pipelines include microGWAS (snakemake pipeline).

Relevant BGWAS Plots



Manhattan plot



Q-Q plot

Interpretation of pyseer output

Column	Meaning	How to interpret
Variant*	Identifier of the genetic variant tested (e.g., k-mer, unitig, gene, SNP).	Indicates which genetic feature was analysed.
Af*	Allele frequency (Number of genomes containing genetic variant/Total number of genomes).	Proportion of isolates carrying the variant.
filter-pvalue	P-value from the preliminary filtering step.	Used internally by pyseer to discard uninformative variants before association testing.
lrt-pvalue*	Likelihood Ratio Test (LRT) p-value (the main value used to determine statistical significance).	Statistical significance of the association after fitting the model.
Beta*	Estimated effect size.	Indicates the strength and direction of the association. Positive (variant increases the phenotype); Negative (variant decreases the phenotype); Near zero (little or no effect).
beta-std-err*	Standard error of the effect size.	Measures the uncertainty of the estimated effect.
intercept	Estimated model intercept.	Baseline phenotype value when the variant is absent.
variant_h2	Estimated proportion of phenotypic variance explained by the variant (when reported).	Larger values suggest a stronger contribution to the phenotype.
Notes	Warnings generated using model fitting	Warns that statistical estimates may be unreliable

Note	Meaning	Interpretation	Recommendation
No note	Model fits successfully	The reported p-value is reliable	Interpret the association normally
bad-chisq	The likelihood ratio test does not follow the expected chi-square distribution.	The reported p-value may not be reliable.	Treat the result with caution; inspect the variant further.
high-bse	High beta standard error.	The estimated effect size (beta) is unstable or imprecise.	The association may be driven by too few isolates or poor model fitting. Could also indicate strong correlation with population structure
bad-chisq, high-bse	The likelihood ratio test does not follow the expected chi-square distribution. High beta standard error.	Both the p-value and effect size are potentially unreliable	Consider the result low confidence unless it is supported by independent analyses or biological evidence.

References

- Lees 2017. The background of bacterial GWAS
https://figshare.com/articles/thesis/The_background_of_bacterial_GWAS/5550037?file=9624721
- Yang et al., 2025. Bacterial genome-wide association studies: exploring the genetic variation underlying bacterial phenotypes.
- Lees et al., 2018. pyseer: a comprehensive tool for microbial pangenome-wide association studies